OBJECT-ORIENTED ANALYSIS & DESIGN · 140703-OOAD

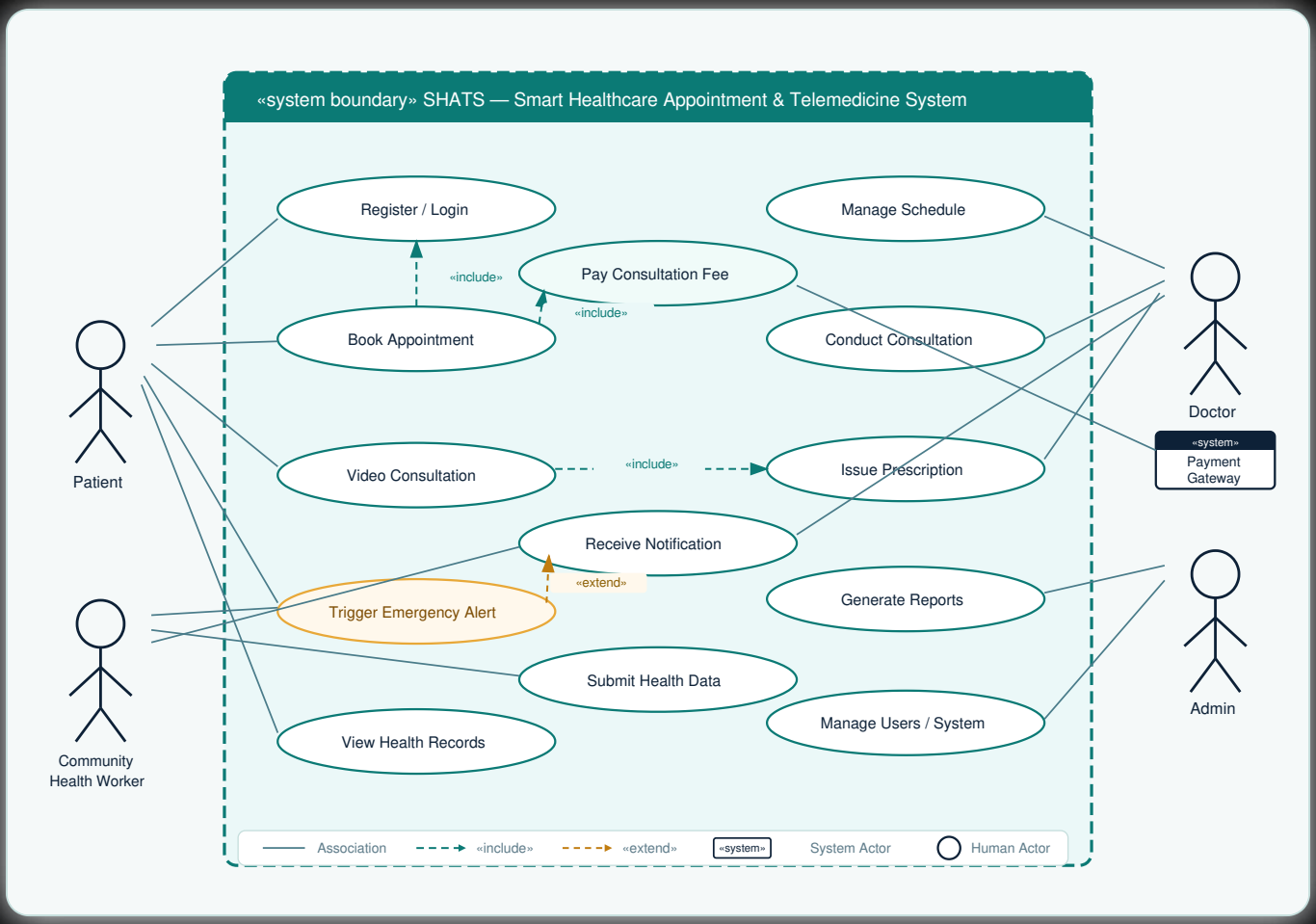
Smart Healthcare Appointment & Telemedicine System

UML Design Documentation — Rural Nigerian Communities · SHATS Platform

1. Use Case Diagram
2. Class Diagram
3. State Diagram
4. Sequence Diagram
5. Activity Diagram

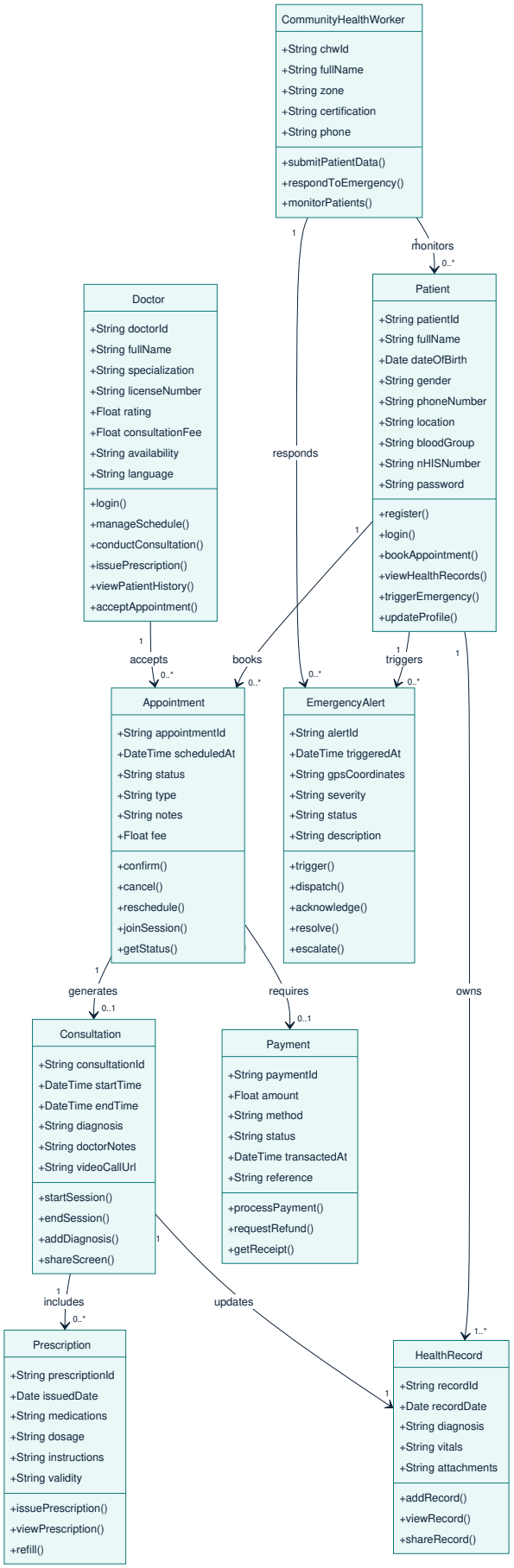
1 Use Case Diagram

System actors and their interactions with the SHATS platform



2 Class Diagram

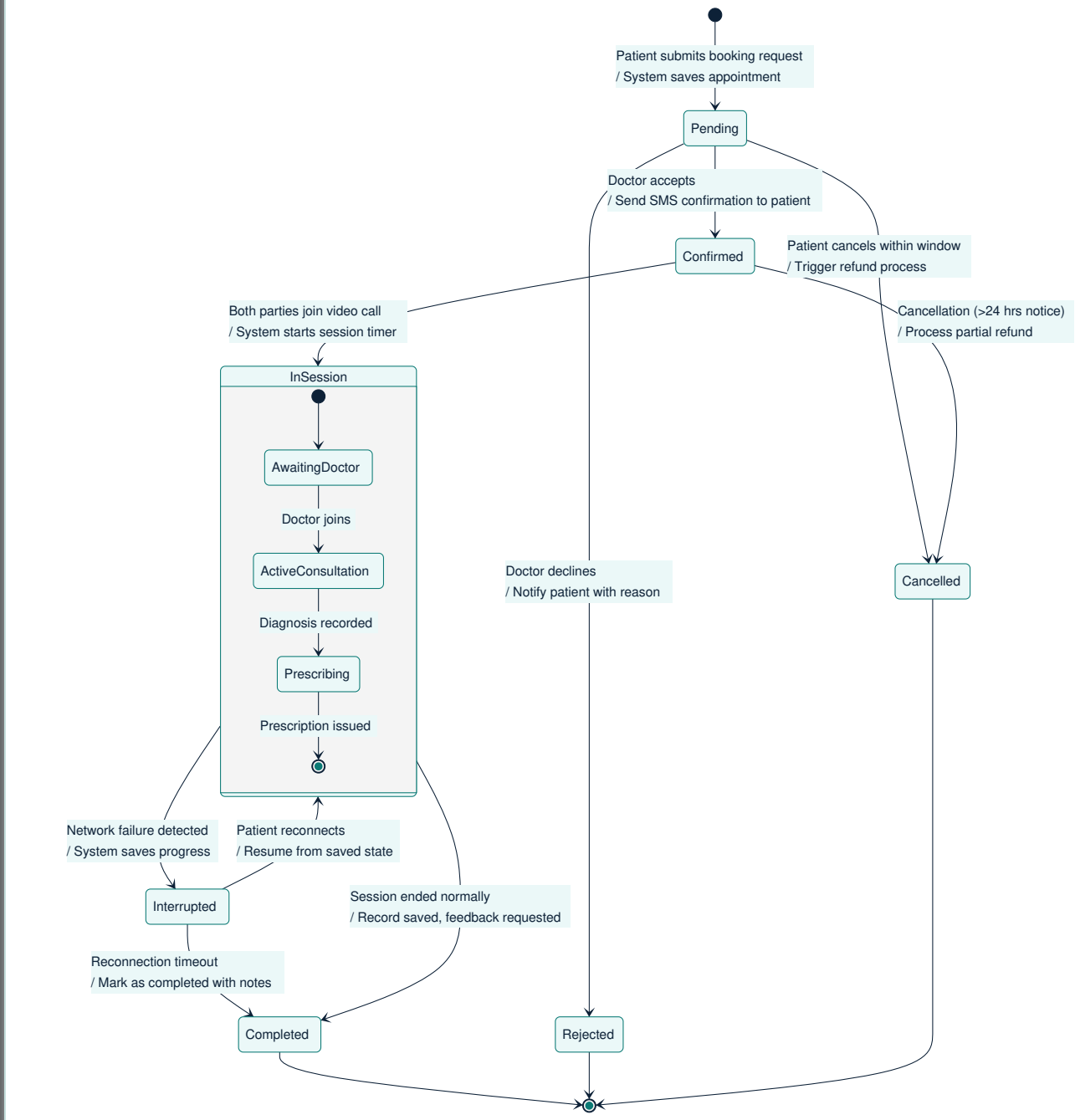
Static structure — classes, attributes, operations and relationships


CommunityHealthWorker
+String chwId
+String fullName
+String zone
+String certification
+String phone
+submitPatientData()
+respondToEmergency()
+monitorPatients()
monitors
Doctor
+String doctorId
+String fullName
+String specialization
+String licenseNumber
+Float rating
+Float consultationFee
+String availability
+String language
+login()
+manageSchedule()
+conductConsultation()
+issuePrescription()
+viewPatientHistory()
+acceptAppointment()
Patient
+String patientId
+String fullName
+Date dateOfBirth
+String gender
+String phoneNumber
+String location
+String bloodGroup
+String nHISNumber
+String password
+register()
+login()
+bookAppointment()
+viewHealthRecords()
+triggerEmergency()
+updateProfile()
responds
accepts
books
triggers
Appointment
+String appointmentId
+DateTime scheduledAt
+String status
+String type
+String notes
+Float fee
+confirm()
+cancel()
+reschedule()
+joinSession()
+getStatus()
EmergencyAlert
+String alertId
+DateTime triggeredAt
+String gpsCoordinates
+String severity
+String status
+String description
+trigger()
+dispatch()
+acknowledge()
+resolve()
+escalate()
generates
requires
owns
Consultation
+String consultationId
+DateTime startTime
+DateTime endTime
+String diagnosis
+String doctorNotes
+String videoCallUrl
+startSession()
+endSession()
+addDiagnosis()
+shareScreen()
Payment
+String paymentId
+Float amount
+String method
+String status
+DateTime transactedAt
+String reference
+processPayment()
+requestRefund()
+getReceipt()
includes
updates
Prescription
+String prescriptionId
+Date issuedDate
+String medications
+String dosage
+String instructions
+String validity
+issuePrescription()
+viewPrescription()
+refill()
HealthRecord
+String recordId
+Date recordDate
+String diagnosis
+String vitals
+String attachments
+addRecord()
+viewRecord()
+shareRecord()

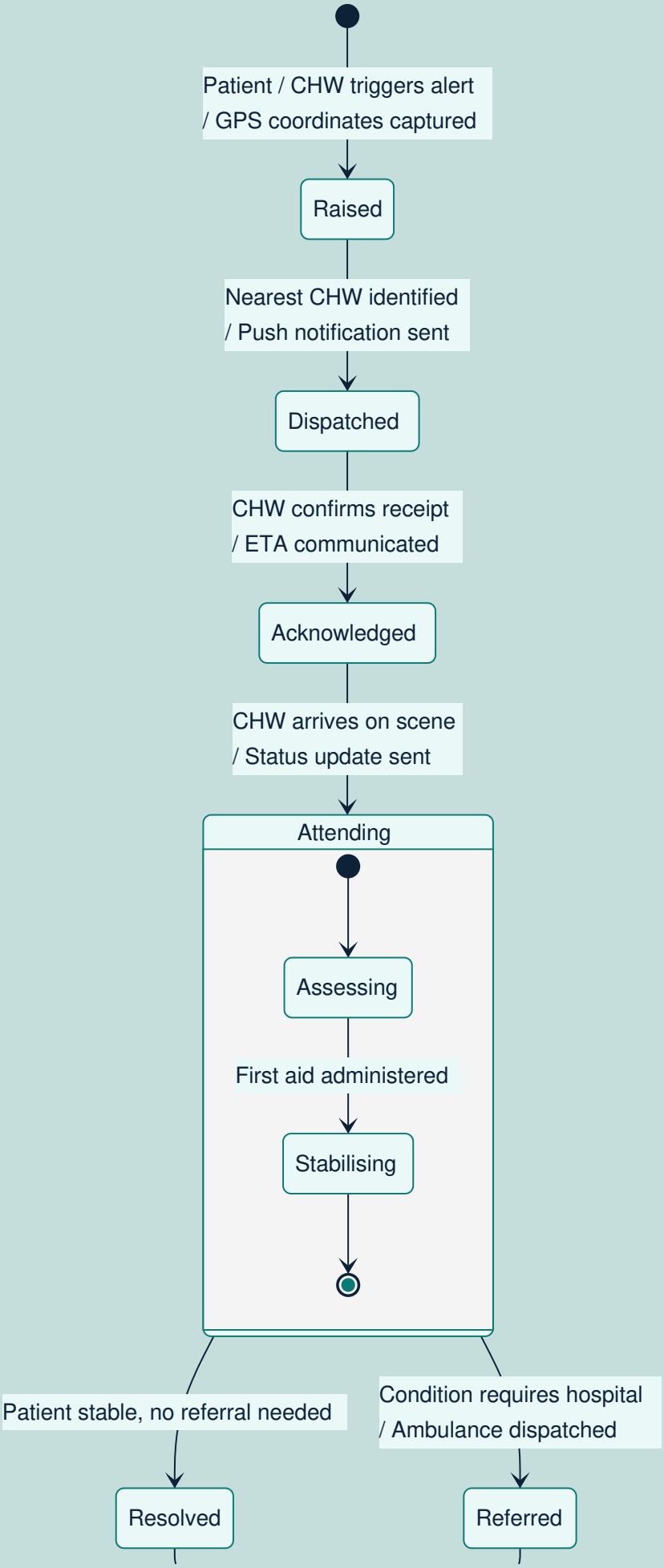
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State Diagram — Appointment Lifecycle

Object state transitions from booking through to completion or cancellation



State Diagram — Emergency Alert Lifecycle

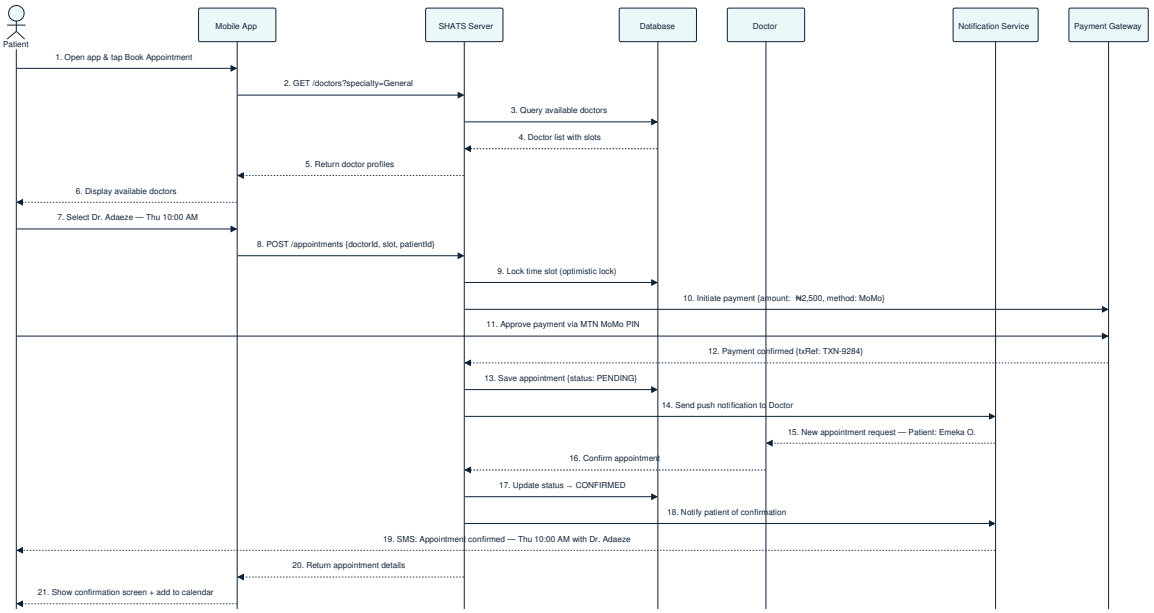

Patient / CHW triggers alert
/ GPS coordinates captured
Raised
Nearest CHW identified
/ Push notification sent
Dispatched
CHW confirms receipt
/ ETA communicated
Acknowledged
CHW arrives on scene
/ Status update sent
Attending
Assessing
First aid administered
Stabilising
Patient stable, no referral needed
Condition requires hospital
/ Ambulance dispatched
Resolved
Referred



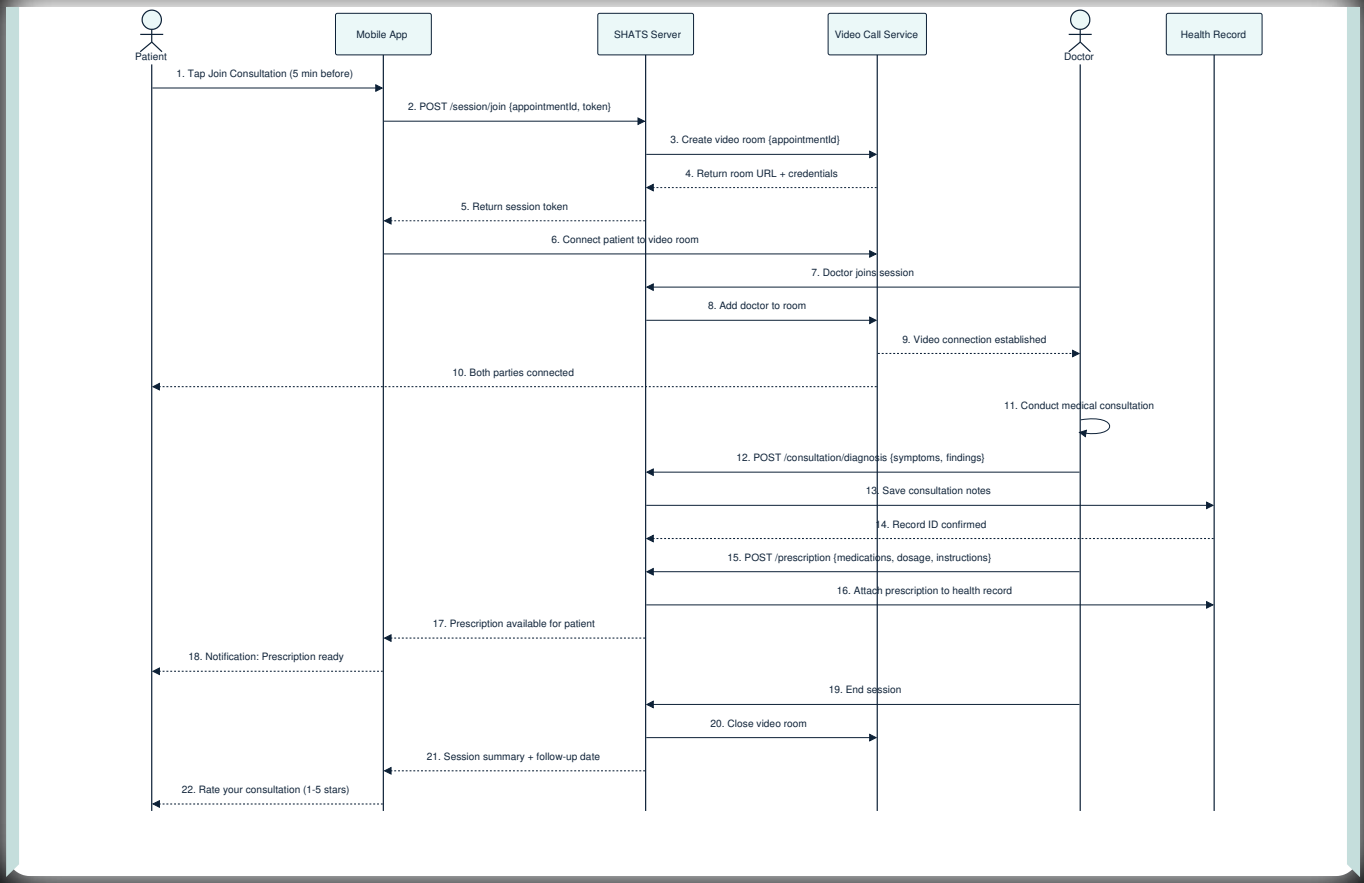
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Sequence Diagram — Book Appointment

Time-ordered message exchanges for the appointment booking workflow



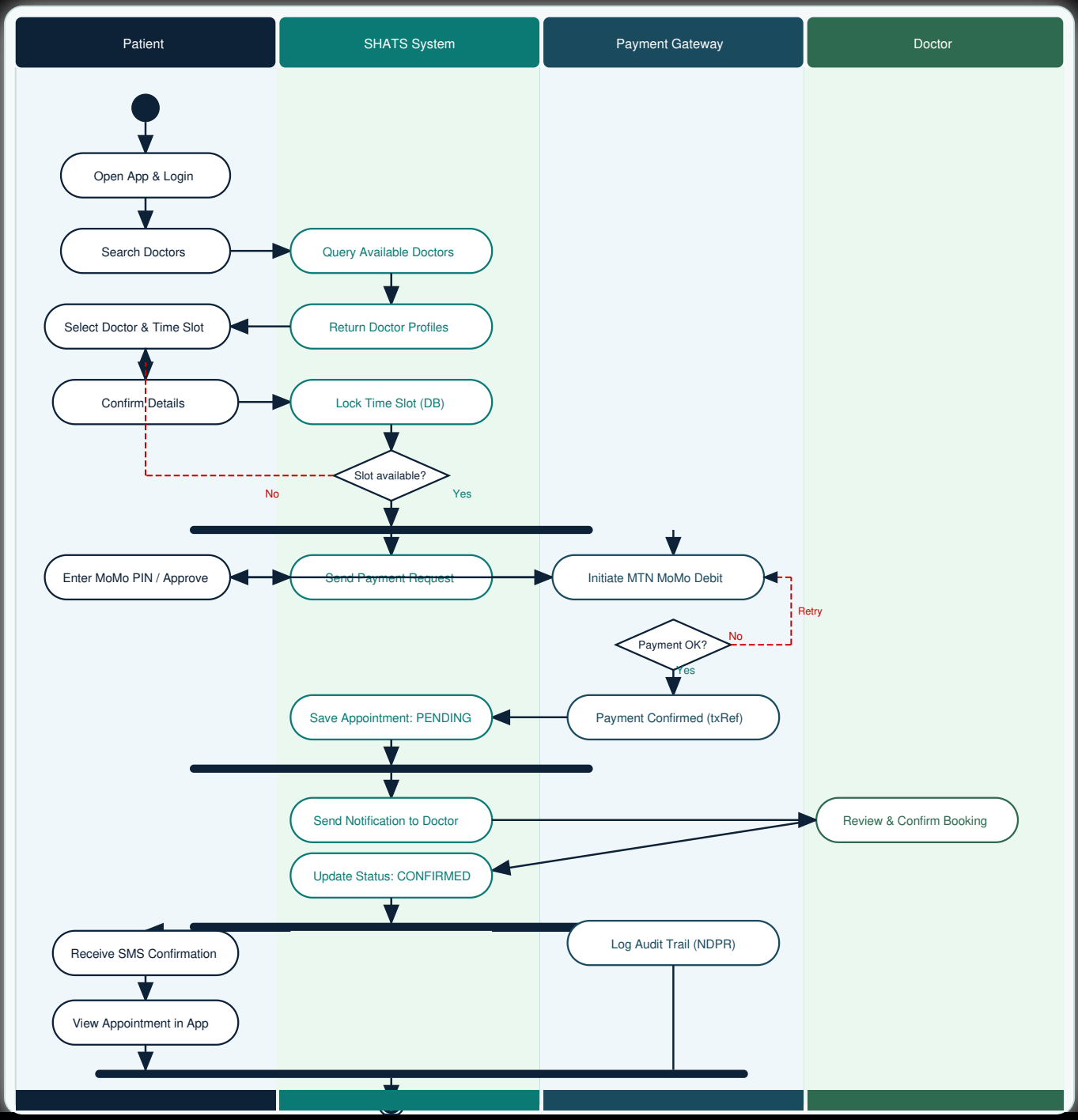
Sequence Diagram — Telemedicine Video Consultation



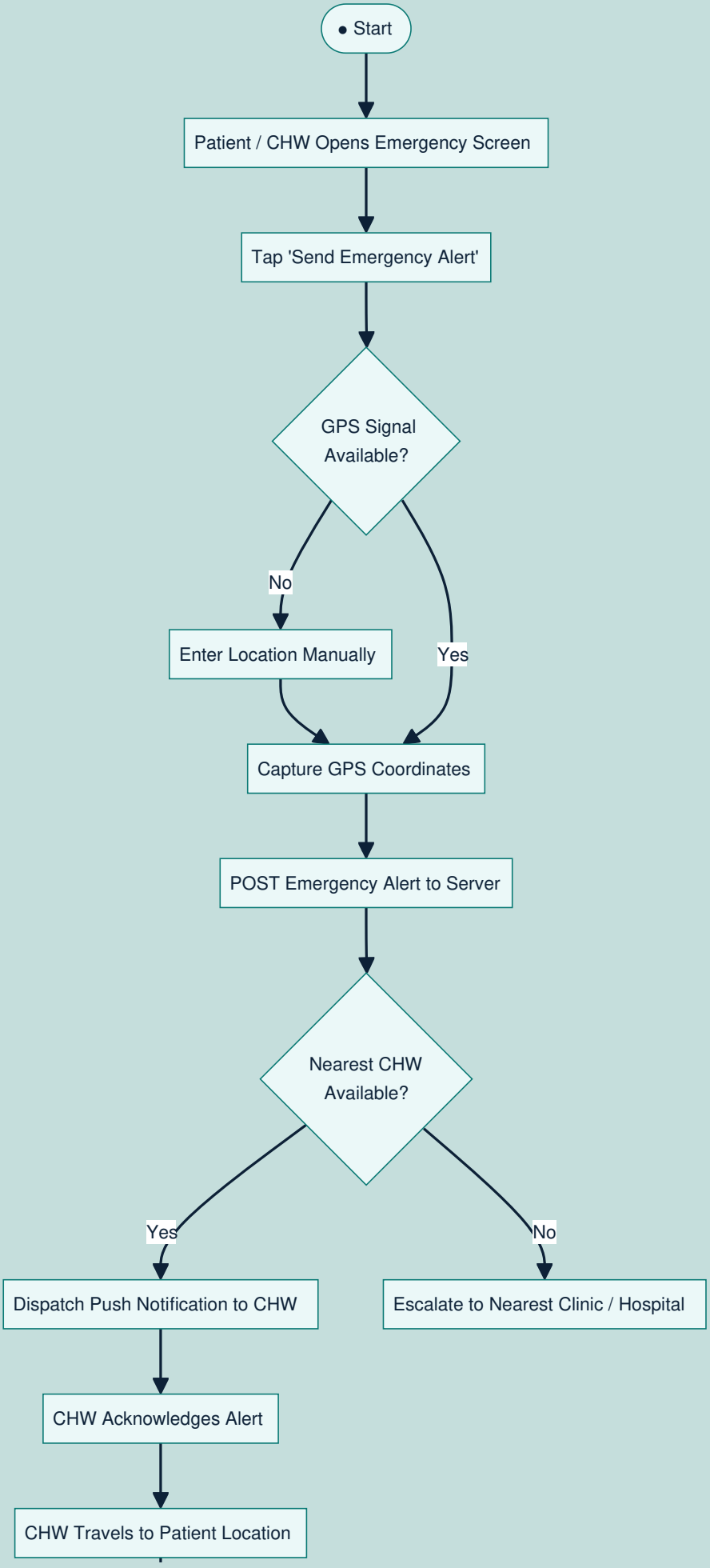
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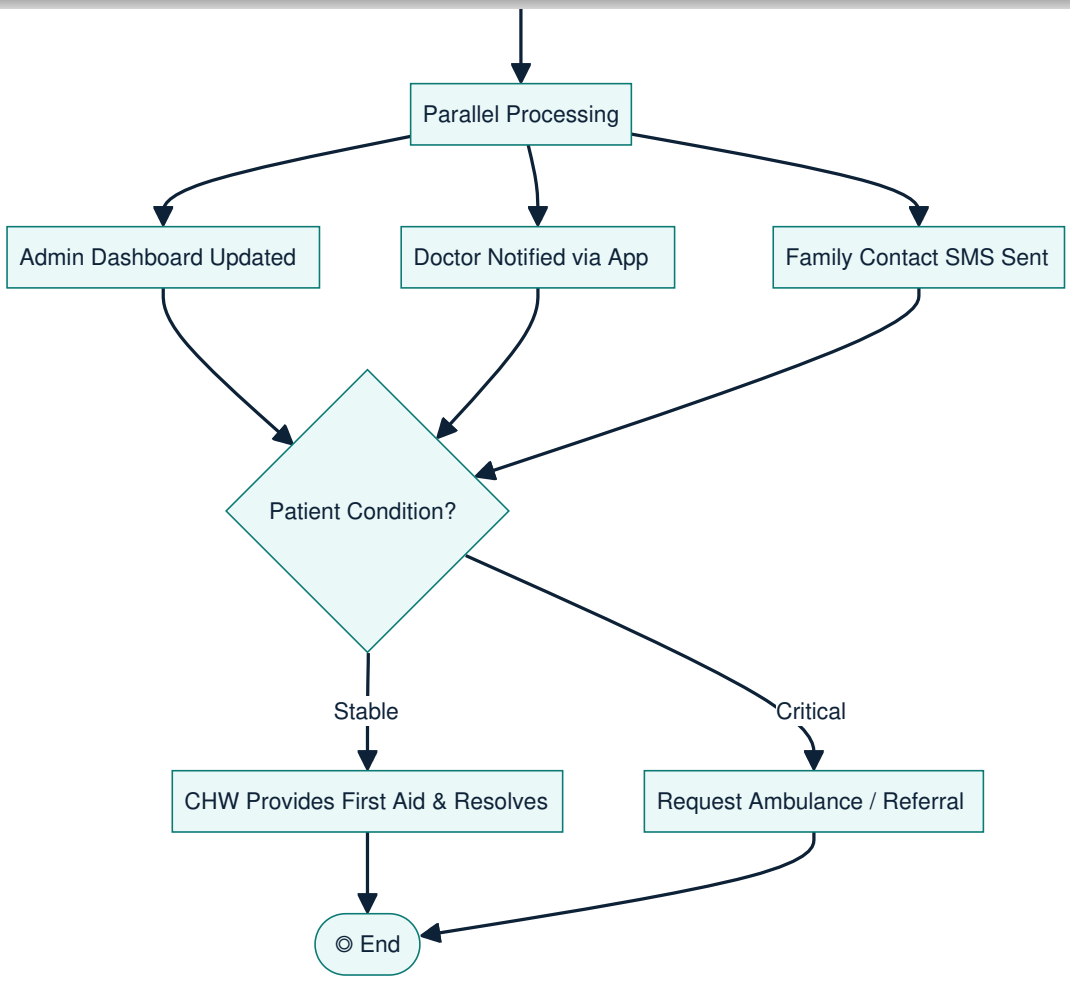
Activity Diagram — Appointment Booking Workflow

End-to-end flow with swimlanes for Patient, System, Doctor, and Payment Gateway



Activity Diagram — Emergency Response Workflow


● Start
Patient / CHW Opens Emergency Screen
Tap 'Send Emergency Alert'
GPS Signal Available?
No
Yes
Enter Location Manually
Capture GPS Coordinates
POST Emergency Alert to Server
Nearest CHW Available?
Yes
No
Dispatch Push Notification to CHW
Escalate to Nearest Clinic / Hospital
CHW Acknowledges Alert
CHW Travels to Patient Location


Parallel Processing
Admin Dashboard Updated
Doctor Notified via App
Family Contact SMS Sent
Patient Condition?
Stable
Critical
CHW Provides First Aid & Resolves
Request Ambulance / Referral
End